Ramraj Velmurugan

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Director, Labs & AI, Komodo Health SF *June 2025 – present*

- Part of the 4-member Engineering Management Team that oversees *all* engineering and product decisions at Komodo.
- Designed, prototyped, and productionalized Marmot, Komodo's Flagship AI offering.
- Created a full AI Agent platform with Agent-/MCP-management, evals and enterprise-grade tenant provisioning.

Engineering Manager, Compute, at Komodo Health SF *Dec 2024 – June 2025*

- Manage 6-member team responsible for Maplab Enterprise workspaces and 1000+ Snowflake accounts.
- Designed and released a highly available underwriting API, with subsecond random access to 10+TB of records.

Engineering Manager, Office of the CTO (Komodo Health Mavens India) *Apr 2023 – Dec 2024*

- Manage the Internal Data Apps team, releasing the no-code analytics tool Genie, and a data overview app, Atlas.
- Guide the data-related architectural decisions across the organization from the Office of the CTO.

Staff Software Engineer at (Komodo Health Mavens India) *Dec 2022 – Mar 2023*

- Help establish org-wide data architecture policy from the Office of the CTO
- Establish and manage the "Internal Apps" team to facilitate internal analytics and solutions applications.

Senior Software Engineer at (Komodo Health Mavens India) *Jan 2022 – Nov 2022*

DS Tech Lead Contractor at Komodo Health NY *Oct 2019 – Jan 2022*

- Spin up the core data interface team from scratch as part of a complete refactor of the organization's technology stack.
- Create and productionalize a cohort-creation API that powers every product at Komodo Health
- Create a SQL generation engine that optimizes and generates complex queries to scan 100+ TB data in seconds.
- Optimize and contribute to fully automated data pipelines to the client's snowflake data warehousing platform.

Data Scientist at Komodo Health NY *Jan 2018 – Oct 2019*

- Worked as a "full stack data scientist" – fully developing a data analytics tool from prototype to forming a team and being tech-lead of commercial product with > $20M ARR.
- Created an internal dashboard allowing visualization of terabytes of data using dc.js and highly optimized Redshift indexes.
- Developed tree-based ML models that perform entity classification and categorization at scale.

Fellow at Insight Data Science, New York *Sep – Dec 2017*

- Created TweetMap.live to geolocate and visualize tweets during natural disasters.
- Collected 18 million tweets about Hurricane Irma, extracted their location from the text and visualized them on a map.

Graduate Research Assistant at UT Southwestern Medical Center, Dallas *2009 – 2017*

- 3D super-resolution microscopy data-processing (ramrajv.com/smlm)
- Multidimensional imaging data visualization and analysis (ramrajv.com/mila)

Projects

Technical Co-founder of inferalpha.com, a book-lending startup *09-2014 – 11-2017*

- Implemented the front- and back-end of a children's books-subscription site using ASP and Azure SQL, hosted using Microsoft's BizSpark program.

Academic software development *08-2009 – 12-2017*

- Worked as part of the UI and software development team for Microscopy Image Analysis Tool (wardoberlab.com/software/miatool), for multidimensional data visualization and processing

Education

***PhD*, Biomedical Engineering** *08-2009 – 08-2017*
The University of Texas Southwestern Medical Center at Dallas, TX.

***Master of Science*, Molecular Biology** *07-2008 – 07-2009*
Institute for Research in Immunology and Cancer, Université de Montréal, Canada.

***Bachelor of Technology,* Industrial Biotechnology** *06-2004 – 04-2008*
AC College of Technology, Anna University, Chennai, India

Publications

	Citations	Year
Microscopy data analysis:		
▪ Sage D, Pham TA, … Velmurugan R, … Holden S. (7th of 20 total authors) Super-resolution fight club: assessment of 2D and 3D single-molecule localization microscopy software. **Nature Methods.**	392	2019
▪ Lin D, Lin Z, Cao J, Velmurugan R, Ober RJ, Ward ES. A two-stage method for automated detection of ring-like endosomes in fluorescent microscopy images. **PloS ONE**.	6	2019
▪ Velmurugan R, Chao J, Ram S, Ward ES, and Ober RJ. Intensity-based axial localization approaches for multifocal plane microscopy. **Optics Express.**	9	2017
▪ Chao J, Velmurugan R, You S, Kim D, Ward ES and Ober RJ. Remote focusing multifocal plane-microscopy for the imaging of 3D single molecule dynamics with cellular context. **Proceedings of SPIE**.	5	2017
▪ Lin D, Lin Z, Velmurugan R, Ober RJ. Automatic endosomal structure detection and localization in fluorescence microscopic images. **IEEE International Symposium on Circuits and Systems**.	1	2017
Cell biology:		
▪ Velmurugan R, Ramakrishnan S, Kim M, Ober RJ, Ward ES. Phagocytosis of antibody-opsonized tumor cells leads to the formation of a discrete vacuolar compartment in macrophages. **Traffic.**	11	2018
▪ Hyenne V, Tremblay-Boudreault T, Velmurugan R, Grant BD, Loerke D, Labbé JC. RAB-5 controls the cortical organization and dynamics of PAR proteins to maintain *C. elegans* early embryonic polarity. **PLoS ONE.**	31	2012
Therapeutic antibody biology:		
▪ Challa DK, Wang X, Montoyo HP, Velmurugan R, Ober RJ, Ward ES. Neonatal Fc receptor expression in macrophages is indispensable for IgG homeostasis. **mAbs**.	63	2019
▪ Khare P, Challa DK, Devanaboyina SC, Velmurugan R, Hughes S, … Ober RJ, Ward ES. Myelin oligodendrocyte glycoprotein-specific antibodies from multiple sclerosis patients exacerbate disease in a humanized mouse model. **Journal of Autoimmunity**.	38	2018
▪ Velmurugan R, Challa DK, Ram S, Ober RJ, Ward ES. Macrophage-mediated trogocytosis leads to death of therapeutic antibody-opsonized tumor cells. **Molecular Cancer Therapeutics**.	115	2016
▪ Li R, Chiguru S, Li L, Kim D, Velmurugan R, Kim D, … Mason R, Ober RJ, Ward ES. Targeting phosphatidylserine with calcium-dependent protein-drug conjugates for the Treatment of Cancer. **Molecular Cancer Therapeutics**.	21	2017
▪ Challa DK, Velmurugan R, Ober RJ, Ward ES. FcRn: From Molecular Interactions to Regulation of IgG Pharmacokinetics and Functions. **Fc Receptors**. *(Book chapter)*	115	2014
▪ Ward ES, Velmurugan R, Ober RJ. Targeting FcRn for therapy: from live cell imaging to in vivo studies in mice. **Immunology Letters.**	16	2014
▪ Bansal P, Khan T, Bussmeyer U, Challa DK, Swiercz R, Velmurugan R, Ober RJ, Ward ES. The encephalitogenic, human myelin oligodendrocyte glycoprotein-induced antibody repertoire is directed toward multiple epitopes in C57BL/6-immunized mice**. Journal of Immunology.**	24	2013